

Date	Activity	Amount
3-Mar	Gasoline	\$26.61
12-Mar	School Trip	\$300.00

6.11 Buying or Leasing a Car



Figure 6.25 The choice to lease or buy is based on cost and other concerns. (credit: “Car Insurance” by Pictures of Money/ Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Evaluate basics of car purchasing.
2. Compute purchase payments and identify the related interest cost.
3. Evaluate the basics of leasing a car.
4. Identify and contrast the pros and cons of purchasing versus leasing a car.
5. Investigate the types of car insurance.
6. Solve application problems involving owning and maintaining a car.

There are people who don't need a car and won't purchase one. But for many people, whether or not to have a car is not a question. Having a car is a basic necessity for these people.

Obtaining a car can be daunting. The models, the features, the additional costs, and finding funding are all steps that need to be taken. One of the big decisions is whether to buy the car or to lease the car. This section will address some of the issues associated with each option.

The Basics of Car Purchasing

The biggest questions you will answer before purchasing a car are, what do you want and what do you need?

Does it have to be new? Does it have to be a make and model you are familiar with? Does it have to have assisted driving? What other details are important to you? For a new vehicle, every feature beyond standard features comes with additional cost, which leads to the question that constrains all of your decisions about a car. How much can you afford to spend on a car?

What you can afford must include insurance costs (discussed later in this section) and maintenance and upkeep. Once you have this in mind, you can search for a car that matches, as closely as possible, what you want and can afford. Most, if not all, dealers have websites that you can search through to identify the car you want. If new cars are not affordable, used cars cost less but come with the wear and tear of use.

The sticker price of the car, called the manufacturer's suggested retail price (MSRP), or the negotiated price you arrive at, isn't the end of the cost to buying a car. There are many fees that accompany the purchase of the car, and perhaps even sales tax. These include but aren't necessarily limited to the following:

- the **title and registration fee**, which includes registering your car with the state, getting the license plate, and assigning the title of the car to the lender. This cannot be avoided.
- a **destination fee**, which covers the cost of delivering the vehicle to the dealer
- a **documentation fee**, sometimes referred to a processing fee of handling fee, is the cost of all the paperwork the dealer did to get you the car
- a **dealer preparation fee**, which is for washing the car and other preparation of that sort. You should try to negotiate that out of the cost of the dealer tries to charge for that
- **extended warranties** and maintenance plans, which help cover some of the costs of caring for the car.
- Sales tax.

You could pay for these immediately, but they are often added to the financing of the car, meaning they become part of the principal of your loan.

EXAMPLE 6.98

Total Cost to Purchase a Car

Nichole negotiates with her car dealership so that the price is \$21,800. She needs to pay the 6.75% sales tax on the car. Other fees are \$31.00 for title and registration, \$1,000 in destination fees, and a documentation fee of \$175. What is the total cost of Nichole's car?

Solution

We add the car's sales cost, sales tax and all other fees to arrive at this value. The sales tax is 6.75% on the price she negotiated, so is $\$21,800 \times 0.0675 = \$1,471.50$. Adding these up, we have a total cost of $\$21,800 + \$1,471.50 + \$31.00 + \$1,000 + \$175 = \$24,477.50$.

YOUR TURN 6.98

1. Luther negotiates for the price of his car, reaching agreement at \$28,975. He needs to pay 8% in sales tax, 2.1% in ownership tax, a \$950 destination fee, processing fees totaling \$370, and registration fee of \$617. What is the total cost for Luther's purchase?

One way to bring down payments on a car is to provide a **down payment** or a trade in. This is money applied to the purchase price before financing happens. Be warned, the sales tax applies to the full purchase price! If you reduce the amount financed, the payments respond by going down. This often becomes part of the negotiating process.

EXAMPLE 6.99

Total Cost to Purchase a Car with Down Payment

Sophia negotiates a \$19,800 price for her new car. The sales tax is 9.5% in her area, and the dealership charges her \$300 in documentation fees. Her title, plates, and registration come to \$321.50. The dealership adds to this a destination fee of \$1,100. If she places a down payment of \$5,000 on the car, what is the total she will finance for the car?

Solution

The price was \$19,800. The sales tax of 9.5% is based on this number. The sales tax comes to $\$19,800 \times 0.095 = \$1,881$. Adding all the fees to the price and the sales tax brings the total cost of the car to $\$19,800 + \$1,881 + \$300 + \$321.50 + \$1,100 = \$23,402.50$. Her down payment is applied to this number, so the \$5,000 is subtracted from \$23,402.50. The subtraction yields the amount to be financed, which is \$18,402.50.

YOUR TURN 6.99

1. Carlos buys a car with a negotiated price of \$36,250. The sales tax in his region is 6.5%. The dealer charges a

\$1,200 destination fee and a \$450 documentation fee. He must pay for title, registration, and license plates, which come to \$21.50. If he has a \$7,500 down payment, how much will he have to finance?

When purchasing a car, the total cost to obtain the car is not the only factor in your monthly price. You will also pay an interest rate for the loan you obtain. The interest rate you will get is dependent on your credit score (see [The Basics of Loans](#)). But you can choose from different lenders. The dealership will likely offer to finance your car loan. Frequently, dealerships offer special financing with very low rates. This is to help move inventory, and may indicate their desire to make sales. This might make negotiating easier. Even if the dealership offers financing, check with your bank or credit union to determine the interest rates they are offering. To reduce your payments, choose the lowest rate you can find.

Purchase Payments and Interest

Whether or not you buy a new car or a used car, if you finance the purchase, you are taking out a loan. The interest rates available for used cars are frequently higher than those for new cars. These loan payments work exactly the same way as other loans do as far as payments are concerned. The payment function comes from [The Basics of Loans](#). The difference between financing a new car or a used car is that financing a new car typically comes with a lower interest rate and a longer term than financing a used car.

FORMULA

The payment, pmt , per period to pay off a loan with beginning principal P is $pmt = \frac{P \times (r/n) \times (1 + r/n)^n \times t}{(1 + r/n)^n \times t - 1}$, where r is the annual interest rate in decimal form, t is the term in years, and n is the number of payments per year (typically, loans are paid monthly making $n = 12$).

Note, payment to lenders is always rounded up to the next penny.

 Often, the formula takes the form $pmt = \frac{P \times (r) \times (1 + r)^n}{(1 + r)^n - 1}$, where r is the interest rate per period (annual rate divided by the number of periods per year), and n is the total number of payments to be made.

EXAMPLE 6.100

New Car Payments

In the following, calculate the monthly payment using the given total to be financed, the interest rate, and the term of the car loan.

1. Total to be financed is \$31,885, interest rate is 2.9%, for 5 years.
2. Total to be financed is \$22,778, interest rate is 4.5%, for 6 years.

Solution

1. The amount to be financed is the principal, P , which is \$31,885. The rate r is 0.029, and the term is $t = 5$ years. These are monthly payments, so $n = 12$. Substituting and calculating, we find the monthly payment to be

$$\begin{aligned}
 pmt &= \frac{P \times (r/n) \times (1 + r/n)^n \times t}{(1 + r/n)^n \times t - 1} \\
 &= \frac{\$31,885 \times (0.029/12) \times (1 + 0.029/12)^{12 \times 5}}{(1 + 0.029/12)^{12 \times 5} - 1} \\
 &= \frac{\$31,885 \times (0.00241\bar{6}) \times (1.00241\bar{6})^{60}}{(1.00241\bar{6})^{60} - 1} \\
 &= \frac{\$31,885 \times (0.00241\bar{6}) \times (1.15583736592)}{1.15583736592 - 1} \\
 &= \frac{\$89.0635298302}{0.15583736592} \\
 &= \$571.52
 \end{aligned}$$

2. The amount to be financed is the principal, P , which is \$22,778. The rate r is 0.045, and the term is $t = 6$ years. These are monthly payments, so $n = 12$. Substituting and calculating, we find the monthly payment to be

$$\begin{aligned}
 pmt &= \frac{P \times (r/n) \times (1 + r/n)^n \times t}{(1 + r/n)^n \times t - 1} \\
 &= \frac{\$22,778 \times (0.045/12) \times (1 + 0.045/12)^{12 \times 6}}{(1 + 0.045/12)^{12 \times 6} - 1} \\
 &= \frac{\$22,778 \times (0.00375) \times (1.00375)^{72}}{(1.00375)^{72} - 1} \\
 &= \frac{\$22,778 \times (0.00375) \times (1.3093031051)}{1.3093031051 - 1} \\
 &= \frac{\$111.837397673}{0.3093031051} \\
 &= \$361.58
 \end{aligned}$$

> YOUR TURN 6.100

In the following, calculate the monthly payment using the given total to be financed, the interest rate, and the term of the car loan.

1. Total to be financed is \$18,325, interest rate is 6.75%, for 4 years.
2. Total to be financed is \$41,633, interest rate is 3.9%, for 6 years.

EXAMPLE 6.101

Used Car Payments

Calculate the monthly payment for the used car if the total to be financed is \$16,990, the interest rate is 7.5%, and the loan term is 3 years.

✓ Solution

The amount to be financed is the principal, P , which is \$16,990. The rate r is 0.075, and the term is $t = 3$ years. These are monthly payments, so $n = 12$. Substituting and calculating, we find the monthly payment to be

$$\begin{aligned}
 pmt &= \frac{P \times (r/n) \times (1 + r/n)^n \times t}{(1 + r/n)^n \times t - 1} \\
 &= \frac{\$16,990 \times (0.075/12) \times (1 + 0.075/12)^{12 \times 3}}{(1 + 0.075/12)^{12 \times 3} - 1} \\
 &= \frac{\$16,990 \times (0.00625) \times (1.00625)^{36}}{(1.00625)^{36} - 1} \\
 &= \frac{\$16,990 \times (0.00625) \times (1.25144613551)}{1.25144613551 - 1} \\
 &= \frac{\$132.887936515}{0.25144613551} \\
 &= \$528.50
 \end{aligned}$$

> YOUR TURN 6.101

1. Calculate the monthly payment for a car loan that has \$21,845 being financed at an interest rate of 6.3% for 4 years.

The Basics of Leasing a Car

Leasing a car is an alternative to purchasing a car. It is still a loan, and acts like one in many respects. They typically last either 24 months or 36 months, though other terms are available. Leases also come with mileage limits, frequently 10,000, 12,000, or 15,000 miles per year. When the lease is over, the car is returned to the dealer. At that time there may be fees that have to be paid, such as for damage to the car or for extra miles driven over the limit.

There are two components to lease costs. One is the monthly payment for the lease. The other is the fees for leasing. These often are paid before the lease is complete. These include:

- a down payment, which is your initial payment that is applied to the price of the car. It reduces the amount you finance, much the same as when you purchase a car. It is recommended that this be negotiated away.
- the **acquisition fee**, sometimes called the bank fee. This is the money charge for the company to set up the lease. It is essentially a paperwork fee. It is not likely that this can be negotiated.
- a **security deposit**, which might be required. It is about the same as 1 month's payment for the lease. The deposit is returned to you if the car is in good shape at the end. This can be negotiated away.
- **disposition fees**, which cover the cost the company will incur when they take your car back and are typically between \$200 and \$450.
- the title, registration, and license fees, just as with the purchase of a car.
- sales tax, which will likely be applied. The sales tax only covers the depreciated portion of the car (more on depreciation later) in many states. Since this depends on the state in which the car is leased, you should determine the sales tax rules for where you lease the car.

As you can imagine, this can come to a fairly high dollar amount.

EXAMPLE 6.102

Cost to Obtain a Lease

Donna wants to lease a Subaru Outback in Eden, New York. Find the total cost of obtaining her lease if there is no down payment, \$175.00 in acquisition fees, a security deposit of \$300.00, \$350.00 in disposition fees, \$102.50 in title and registration fees, and sales tax of \$3,536.05.

Solution

Adding these values together we find the total cost is \$4,463.55.

YOUR TURN 6.102

1. RJ wants to lease a Tahoe in Salt Lake City, Utah. Find the total cost of obtaining his lease if he has a \$5,000.00 down payment, \$225 in acquisition fees, a security deposit of \$750.00, \$500.00 in disposition fees, and sales tax of \$4,092.



PEOPLE IN MATHEMATICS

Zollie Frank

Zollie Frank and Armund Shoen founded one of the original leasing companies, Four Wheels, in 1939. Their company leased automobiles to corporations. They began by leasing 5 cars to the Petrolager pharmaceutical company in year one. This saved Petrolager money and provided a steady cash flow to the Four Wheels business. In year two, the number of cars leased to Petrolager was 75. Their new idea was to lease cars directly to companies for one year. Previously, such companies might pay for mileage, gas, and a partial down payment. Sadly, the salesmen who were being so helped often left the company before the car was paid for, and so the company lost the down payment money.

The lease was for \$45 per month per car for one year.

You have some obligations when you lease a car. You must keep the car in good condition, cleaned, maintained, and free of anything more than minor damage. If the car is in poor condition when the car is returned, you will be responsible for the cost to bring the car to an acceptable condition. You are also expected to keep the mileage under its limit. If you go over, you will pay 10 to 25 cents per mile over.

Lease Payments

Lease payments are similar to regular loan payments, but have some other details. Calculating a lease payment involves knowing the following values:

- **The price of the car.** This is the cost you would pay for the car after applying all discounts, incentives, and

negotiations.

- **Residual Value.** This is the manufacturer's estimate of the car's value after a set period of time. The residual value is expressed as a percentage of the manufacturer's suggested retail price (MSRP).
- **Months.** This is the length of the lease. Most leases are either 24- or 36-month leases, but other terms are available.
- **Monthly Depreciation.** The monthly depreciation is the difference between the price of the car and the residual value, divided by the number of months of the lease, and represents the monthly loss of value of the car while it's being used.
- **Money Factor (MF).** This is the interest rate, but expressed in a different way for a lease. Converting from the money factor to the annual percentage rate (APR) is done by multiplying the MF by 2400. Naturally, converting an APR to a MF is done by dividing the APR by 2400.

FORMULA

The monthly depreciation for a car, MD, is $MD = \frac{P-R}{n}$, P is the price paid for the car, R is the residual value of the car, and n is the number of months of the lease.

The annual percentage rate for a lease is $APR = 2400 \times MF$, where MF is the money factor of the lease. The MF for a lease is $MF = APR/2,400$.

EXAMPLE 6.103

Monthly Depreciation of a Car

1. The purchase price of a car is \$25,000. Its residual price is \$14,500. What is its monthly depreciation for a 36-month lease?
2. The purchase price of a car is \$30,000. Its residual price is \$18,600. What is its monthly depreciation for a 24-month lease?

Solution

1. The monthly depreciation formula is $MD = \frac{P-R}{n}$. Substituting \$25,000 for P , \$14,500 for R , and 36 for n , we find MD to be $MD = \frac{P-R}{n} = \frac{\$25,000 - \$14,500}{36} = \frac{\$10,500}{36} = \$291.67$.
2. The monthly depreciation formula is $MD = \frac{P-R}{n}$. Substituting \$30,000 for P , \$18,600 for R , and 24 for n , we find MD to be $MD = \frac{P-R}{n} = \frac{\$30,000 - \$18,600}{24} = \frac{\$11,400}{24} = \$475.00$.

YOUR TURN 6.103

1. The purchase price of a car is \$27,500. Its residual price is \$17,875.00. What is its monthly depreciation for a 24-month lease?

EXAMPLE 6.104

Converting Between APR and MF

1. Find the annual percentage rate if the money factor is 0.00001875.
2. Find the money factor if the APR is 6.25%.

Solution

1. The APR is the money factor times 2400, so $APR = 2400 \times MF = 2400 \times 0.00001875 = 0.045$. Expressed as a percentage, the APR is 4.5%.
2. The MF is the APR divided by 2400, so $MF = APR/2400 = 0.0625/2400 = 0.000026041\bar{7}$.

YOUR TURN 6.104

1. Find the annual percentage rate if the money factor is $0.00001208\bar{3}$.

2. Find the money factor if the APR is 7.9%.

Once the values above are found, the payment for the lease can be calculated.

FORMULA

The payment, PMT , for a lease is $PMT = \frac{(P - R)}{n} + (P + R) \times MF$, where P is the price paid for the car, R is the residual value of the car, n is the number of months of the lease, and MF is the money factor for the lease.

EXAMPLE 6.105

Calculating Car Lease Payments

Calculate the lease payments for car with the following price, residual price, length of lease, and money factor or APR.

1. Price is \$28,344, residual price is \$18,140.16, 24-month lease, money factor is 0.000025.
2. Price is \$22,500, residual price is \$13,050, 36-month lease, APR is 7.5%.

Solution

1. Substituting the values $P = \$28,344$, $R = \$18,140.16$, $n = 24$ and $MF = 0.000025$ into the formula and calculating, the monthly lease payment is

$$\begin{aligned} PMT &= \frac{(P - R)}{n} + (P + R) \times MF \\ &= \frac{(\$28,344 - \$18,140.16)}{24} + (\$28,344 + \$18,140.16) \times 0.000025 \\ &= \frac{\$10,203.84}{24} + (\$46,484.16) \times 0.000025 \\ &= \$426.33 \end{aligned}$$

2. Given the APR, we find the MF which is $MF = APR/2400 = 0.075/2400 = 0.00003125$. Substituting the values $P = \$28,344$, $R = \$18,140.16$, $n = 24$ and the MF into the formula and calculating, the monthly lease payment is

$$\begin{aligned} PMT &= \frac{(P - R)}{n} + (P + R) \times MF \\ &= \frac{(\$22,500 - \$13,050)}{36} + (\$22,500 + \$13,050) \times 0.00003125 \\ &= \frac{\$9,450}{36} + (\$35,550) \times 0.000025 \\ &= \$263.62 \end{aligned}$$

YOUR TURN 6.105

Calculate the lease payments for car with the following price, residual price, length of lease, and money factor or APR.

1. Price is \$38,750, residual price is \$18,140.16, 36-month lease, money factor is 0.000035.
2. Price is \$45,600, residual price is \$21,312.50, 24-month lease, APR is 11.7%.

Comparing Purchasing and Leasing

When deciding to buy or lease a car, the differences between the two options should be carefully evaluated. The following is a list of points of comparison between the two.

- The payments for a lease are likely less than the payment for purchasing.
- When leasing, you get a new car after the lease term is over, typically 24 or 36 months. Buying the car means the same car is driven until it is re-sold and a new one bought. Essentially, leasing a car is equivalent to renting a car.
- The leased car is new, so all warranties are in force and you drive the car during its best years. When the car is purchased, it may be kept past its warranties and may be driven until it is quite old.
- Each time you lease a new car, all the fees and taxes must be paid again. When buying a car, these fees are only paid once.
- Leasing contracts carry restrictions on the mileage you can drive per year, and going over incurs more cost at the

end of the lease. Buying the car means no such mileage limits.

- When leasing, you are obligated to keep the vehicle in good condition and maintained according to the dealer's schedule. Some dealerships will even pay for oil changes over the life of the lease. When the car is purchased, the upkeep schedule is the choice of the owner.
- When a car is purchased and kept for long enough, the warranty expires and the owner is responsible for all maintenance items and repairs. The warranty for a car won't expire during the lease term.
- When a new car is purchased and the loan is paid off the car is still owned by the buyer and may be traded in when a new car is to be purchased. When leasing, the car is returned to the dealer when the lease term is over.

When deciding between the two, you are choosing between these features. If you aren't willing to drive an older car or deal with the upkeep that accompanies an older car, you may want to lease. This means you will need to pay those beginning costs each time the lease is up. If you want to own the car after the payments are over, then you may want to buy a car. This means you are paying for all the upkeep after the warranties expire, but you have no limits on mileage and own the car at the end. It really depends on your preferences.

EXAMPLE 6.106

Lease or Buy

In the following, determine if a lease or purchase of a car is better.

1. Joyce is concerned with large repairs and does not want to deal with them.
2. Maurice prefers to drive newer cars.

Solution

1. Since Joyce does not want to deal with repairs, so leasing would be a better choice. This way, the warranty covers most of the big repairs that could need to be done.
2. Since Maurice likes to drive new cars, leasing is a better option, since he will lease a new car every 2 to 3 years.

YOUR TURN 6.106

1. Christopher does a lot of driving, averaging over 20,000 miles per year.
2. John eventually wants to not make car payments.

Car Insurance

Car insurance is meant to cover costs associated with accidents involving cars. Most states (all except New Hampshire and Virginia) require some insurance. Without insurance, the state may not let you get a license for your car or register your car. Your state's requirements can be hard to follow. Fortunately, insurance companies and brokers will make sure your insurance is sufficient for your state and will warn you if you try to not meet the requirements. Of course, they may offer more than what is sufficient, so it is your responsibility to determine how much coverage you want, as long as the minimum insurance requirements are met. The cost of insurance should be accounted for when evaluating the affordability of buying or leasing a car.

Whether your car is leased or owned, you do need insurance. This contributes to the cost of having the vehicle. Leasing or owning makes no difference to the insurance company you choose, because they are insuring you based on what you are driving, your driving record, and other information about you including where you live and your age. These insurance policies have many components that address different costs that can come from auto accidents. This may make details confusing, and you may not realize what you are paying for until you must use it. Here is a brief outline of the different components of auto insurance, many of which are required by the state that issues your driver's license.

- **Liability insurance** is mandatory coverage in most states. Liability insurance covers property damage and injuries to others should you be found legally responsible for an accident. You are required to have the minimum amount of coverage, as determined by your state, in both areas.
- **Collision insurance** is insurance covering the damage caused to your car if involved in an accident with another vehicle.
- **Comprehensive insurance** is an extra level of coverage if involved in an accident with another vehicle and covers other things like theft, vandalism, fire, or weather events as outlined in your policy. There is a deductible assigned to each type of insurance, an amount that you pay out of pocket before your comprehensive coverage takes effect. Comprehensive insurance is often required if you lease or finance the purchase of a vehicle.

- **Uninsured or underinsured motorist insurance:** If you are hit by an uninsured or underinsured motorist, this insurance will help pay medical bills and damage to your car.
- **Medical payments insurance** is mandatory in some states and helps pay for medical costs associated with an accident, regardless of who is at fault.
- **Personal injury protection insurance** is coverage for certain medical bills and other expenses due to a car accident. Other covered expenses may include loss of income or childcare, depending on your policy.
- **Gap insurance** is designed to cover the gap between what is owed on the car and what the car is worth in the event your car is a total loss.
- **Rental reimbursement insurance** is coverage for a rental car while your car is under repair resulting from an accident.

You can also purchase other special insurance policies, such as classic car insurance, new car replacement insurance, and sound system replacement insurance, to name a few. It is important that you determine exactly what you need, as insurance policies can be expensive and vary according to your age, driving history, and where you live.

EXAMPLE 6.107

Types of Insurance

1. Which component of insurance pays if you are in an accident with a motorist without insurance?
2. Which component of insurance pays for the remaining principal owed on your car in the case of a total loss?

Solution

1. Uninsured motorist insurance covers accidents with those who have no insurance.
2. Gap insurance will cover the gap between what is owed on the car and what it is worth if an accident results in a total loss.

YOUR TURN 6.107

1. Which component of insurance covers medical bills in the event of an accident?
2. Which component of insurance pays for costs to others if you have an accident and are found legally responsible for those damages?

EXAMPLE 6.108

Monthly Cost of Owning a Car

If your car payment is \$287.50 per month and your car insurance is \$930 every 6 months, what is the cost of the car per month when accounting for the insurance?

Solution

The cost of the car including insurance is the monthly payment, \$287.50, plus the monthly cost of the insurance. The insurance cost per month is $\$930/6 = \155 since the insurance cost is for every 6 months. Adding those the cost with insurance is \$442.50.

YOUR TURN 6.108

1. If your car payment is \$410.86 per month and your car insurance is \$2,190 per year, what is the cost of the car per month when accounting for the insurance?

Maintaining a Car

Cars are not a buy it and forget it item. They require upkeep, which adds to the cost of owning the car. Tires, brakes, and wipers need replacing. Oil changes, inspections, so many things other than gasoline. Below is a list of some maintenance requirements for cars, along with cost and roughly how often they should happen.

Maintenance	Frequency	Cost Range
New Tires	Every 5 years	\$25–\$300 per tire
Oil Change	Every 3,000–6,000 miles	\$35–\$75
Wipers	Every 6–12 months	\$20–\$40
Inspection	Annual	\$10–\$50
Brake pads	10,000–20,000 miles	\$200–\$300
Air Filter	15,000–30,000 miles	\$35–\$80

When designing a budget, these expected costs should be accounted for. Extra money per month should be saved in addition to this budget category, to handle unanticipated, and perhaps very costly, repairs.

EXAMPLE 6.109

Estella needs to budget for her car maintenance. She expects to buy new tires each 4 years, which will cost her \$480 to replace them all. Oil changes near her cost \$49.99, and she believes she will get one every 4 months. Her inspection costs \$15 per year. Wipers for her car cost \$95 for all three and she anticipates changing them every year. She drives less than 30,000 miles per year, so she plans to replace the air filter once per year. The air filter for her car costs \$57.50. How much should she budget per month to cover these costs?

✓ Solution

Her yearly costs are the wipers, inspection, and tires, which total \$167.50. Tires will be bought every 5 years, so per year she should budget \$96. Her oil changes, which will happen three times per year, cost \$49.99 each, so she'll spend \$149.97 for the year on oil changes. Adding these up, her yearly budget should include \$413.47 for maintenance. Dividing by 12 gives the monthly budget for maintenance, which is \$34.46 (rounded up to the next penny).

> YOUR TURN 6.109

1. Natalie needs to budget for her car maintenance. She expects to buy new tires each 5 years, which will cost her \$390 to replace them all. Oil changes near her cost \$59.99, and she believes she will get one every 3 months. Her inspection costs \$25 per year. Wipers for her car cost \$115 for all three and she anticipates changing them every year. She drives less than 30,000 miles per year, so she plans to replace the air filter once per year. The air filter for her car costs \$46.25. How much should she budget per month to cover these costs?

Check Your Understanding

69. What is destination fee?
70. What is a title and registration fee?
71. Calculate the monthly payment if the total to be financed is \$34,570, the interest rate is 3.5%, for 5 years.
72. What might be different for a used car loan and a new car loan?
73. What fees are similar between leasing a car and buying a car?
74. If the cost of a car is \$30,000, and the residual value of the car after 3 years is \$18,000, what is the monthly depreciation for the car?
75. If the money factor for a lease is 0.000015625, what is the annual interest rate?
76. For people who do not mind driving an older car or maintaining a car, which is preferable, a lease or purchasing?

77. What does collision insurance cover?



SECTION 6.11 EXERCISES

1. What are the two main questions to answer before buying a car?
2. What is a documentation fee?
3. What is a dealer preparation fee?
4. What is a down payment?
5. How long do leases typically last?
6. Can you drive a leased car any mileage?
7. What are three fees associated with leasing a car?
8. What is the monthly depreciation for a leased car?
9. How are the money factor and the annual percentage rate related?
10. Name two advantages to leasing over buying a car.
11. Name two advantages to buying a car over leasing a car.
12. What does liability insurance cover?
13. What does collision insurance cover?
14. Many expenses associated with a car can be anticipated. Name three maintenance expenses that can be anticipated.

In the following exercises, find the total cost to purchase the car.

15. Alexia negotiates a purchase price of \$17,850 for her new car. The sales tax in her area is 6.5%. Her license, plates, and registration come to \$285.00. The dealership charges her a \$600 destination fee and a \$150 processing fee. How much will she finance in total for the car?
16. Stephanie negotiates a purchase price of \$25,670 for her new car. The sales tax in her area is 8.0%. Her license, plates, and registration come to \$389.00. The dealership charges her a \$700 destination fee and a \$345 processing fee. How much will she finance in total for the car?
17. Matthew negotiates a purchase price of \$35,100 for her new car. The sales tax in his area is 7.25%. His license, plates, and registration come to \$325.00. The dealership charges him a \$900 destination fee and a \$125 processing fee. How much will he finance in total for the car?
18. Madisyn negotiates a purchase price of \$45,800 for her new car. The sales tax in her area is 7.25%. Her license, plates, and registration come to \$199.00. The dealership charges her a \$1,000 destination fee and a \$275 processing fee. How much will she finance in total for the car?

In the following exercises, calculate the car payment based on the total financed and the interest rate.

19. Total to be financed is \$36,775, interest rate is 2.75%, for 6 years.
20. Total to be financed is \$29,350, interest rate is 3.9%, for 5 years.
21. Total to be financed is \$27,180, interest rate is 1.99%, for 7 years.
22. Total to be financed is \$15,489, interest rate is 6.75%, for 4 years.

In the following exercises, find the total cost to obtain the lease.

23. A \$2,000.00 down payment, \$120 in acquisition fees, a security deposit of \$350.00, \$200.00 in disposition fees, and sales tax of \$2,860.
24. No down payment, \$260 in acquisition fees, a security deposit of \$450.00, \$400.00 in disposition fees, and sales tax of \$3,155.00.
25. A \$4,000 down payment, \$360 in acquisition fees, a security deposit of \$900.00, \$1,000.00 in disposition fees, and sales tax of \$4,275.
26. A \$7,000.00 down payment, \$225 in acquisition fees, a security deposit of \$800.00, \$675.00 in disposition fees, and sales tax of \$3,673.

In the following exercises, find the monthly depreciation of the car.

27. The purchase price of a car is \$34,000. Its residual price is \$23,500. What is its monthly depreciation for a 24-month lease?

28. The purchase price of a car is \$23,500. Its residual price is \$11,750. What is its monthly depreciation for a 36-month lease?

In the following exercises find the APR based on the MF.

29. $MF = 0.00004125$
 30. $MF = 0.000046458\overline{3}$

In the following exercises find the MF based on the APR.

31. 8.75%
 32. 6.25%

In the following exercises, find the lease payment based on the given information.

33. Price is \$41,700, residual price is \$27,105, 24-month lease, money factor is 0.000025.
 34. Price is \$22,165, residual price is \$12,855.70, 24-month lease, money factor is 0.0000275.
 35. Price is \$30,650, residual price is \$16,857.50, 36-month lease, APR is 8.25%.
 36. Price is \$24,800, residual price is \$14,384, 36-month lease, APR is 5.85%.
37. Sara needs to budget for her car maintenance. She expects to buy new tires each 3 years, which will cost her \$540 to replace them all. Oil changes near her cost \$39.99, and she believes she will get one every 4 months. Her inspection costs \$20 per year. Wipers for her car cost \$115 for all three and she anticipates changing them every year. She drives less than 30,000 miles per year, so she plans to replace the air filter once per year. The air filter for her car costs \$36.25. She plans to get new brake pads every year, which cost her \$267. How much should she budget per month to cover these costs?
38. Mwibeleca needs to budget for his car maintenance. He expects to buy new tires each 3 years, which will cost her \$560 to replace them all. Oil changes near his cost \$69.99, and he believes he will get one every 4 months. His inspection costs \$25 per year. Brake pads will cost \$215 each year. Wipers for his car cost \$143.50 for all three and he anticipates changing them every year. He drives less than 30,000 miles per year, so he plans to replace the air filter once per year. The air filter for his car costs \$62.88. How much should Mwibeleca budget per month to cover these costs?
39. John will either lease or buy a car. The total cost to purchase the car is \$35,830, and he would finance the car for 5 years at 2.99%. If he leases, he would pay \$3,287.50 for the lease, and then his payments would be based on a price of \$32,750, a residual price of \$20,305, 36 months, with a money factor of 0.00001725. Compare the payments to purchase the car to the payments of the lease plus the lease cost divided by 36.
40. Zachary will either lease or buy a car. The total cost to purchase the car is \$22,945, and he would finance the car for 6 years at 1.99%. If he leases, he would pay \$2,387.75 for the lease, and then his payments would be based on a price of \$21,350, a residual price of \$12,390.30, 36 months, with a money factor of 0.000018375. Compare the payments to purchase the car to the payments of the lease plus the lease cost divided by 36.